Hamed Hojatian

Ph.D. Candidate

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Education

Fall 2019 – Now
Polytechnique Montréal
Ph.D. in AI-aided wireless communication systems
GPA: 4.0/4.0

2014-2017
Isfahan University of Technology (IUT)
M.Sc. in communication system engineering
GPA: 3.7/4.0

2010-2014
Babol (Noshirvani) University of technology (NIT)
B.Sc. in electrical engineering
GPA: 3.1/4.0

Research Experience

Ph.D.
AI-aided Beamforming in massive MIMO Systems
- RSSI-Based hybrid beamforming design with deep learning;
- Improve the spectral and energy efficiency of hybrid beamforming design;
- Unsupervised learning for hybrid beamforming design and intelligent beamforming;
- Coordinated hybrid beamforming and cell-free massive MIMO with deep neural network; and
- Subarray hybrid beamforming and efficiecnt hardware design driven by unsupervised learning.
- ...

M.Sc.
Time and Frequency synchronization in OFDM-based Massive MIMO systems
- Time and frequency synchronization in OFDM-based massive MIMO;
- Carrier Frequency Offset (CFO) and channel estimation;
- Detection in receiver using iterative method; and
- Improving the performance by using Filtered-OFDM.

B.Sc.
OFDM Transmitter and receiver simulation
- Developed in MATLAB simulation
- OFDMA multi-user simulation in AWGN channel.

Skills

Tools and Languages | Python, Pytorch, TensorFlow, MATLAB, HTML and CSS, JavaScript, Bash, and Microsoft office.

Telecomm. Tools | Ericsson OSS, MapInfo, Atoll, TEMS, Actix

Languages | English, French (basic), Persian (native)

Community Service

IEEE Journals reviewer	Selected Areas in Communications (**J-SAC**), Transactions on Wireless Communications (**TWC**), Transactions on Communications (**TCOMM**), Wireless Communications Letters (**WCL**), Communications Letters (**CommL**), and Transactions on Vehicular Technology (**TVT**). Details in here.

Research Interests

- AI-aided wireless communication;
- Machine Learning, Deep Learning, DRL, Federated Learning, META and few shot learning;
- Signal processing in massive MIMO communication, 5G beyond and 6G;
- Intelligent beamforming, coordinated, cooperative beamforming and distributed antenna;
- Millimeter wave in Massive MIMO and Antenna array;
- Statistical learning, optimization; and
- Cell free communication.

Work Experience

Nov 2015 - Nov 2018

Mobile Telecommunication Company of Iran (MCI)
Radio planning and optimization engineer

- Live network Ericsson OSS optimization and parameter tuning for 2G, 3G, and LTE;
- Network Planning (2G, 3G and LTE), new site baseline parameters tuning and planning;
- Project manager of RNC rehoming and RNC planning; and
- Project manager of site renaming.

Oct 2015

Eleventh (11th) national robotics competition at BABOL University of Technology
Vice-President

Publications

2022

1. **Hojatian, H.**, Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Decentralized Beamforming for Cell-Free Massive MIMO with Unsupervised Learning**". ***IEEE Communications Letters*** (2022).
2. **Hojatian, H.**, Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Flexible Unsupervised Learning for Massive MIMO Subarray Hybrid Beamforming**". ***IEEE GLOBECOM*** (2022).

2021

3. **Hojatian, H.**, Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Unsupervised Deep Learning for Massive MIMO Hybrid Beamforming**". ***IEEE Transactions on Wireless Communications*** (2021).

2020

4. **Hojatian, H.**, Nguyen Ha, V., Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**RSSI-Based Hybrid Beamforming Design with Deep Learning**". ***IEEE ICC*** (2020).

2016

5. **Hojatian, H.**, Omidi, M. J., Saeedi-Sourck, H. & Farhang, A. "**Joint CFO and Channel Estimation in OFDM-based Massive MIMO Systems**". ***IEEE IST*** (2016).

Presentation and Teaching Experience

2021 | Presenting my research progress in **IVADO - Digital October.**

2020 | Paper Presentation, **IEEE ICC 2020**, Dublin, Ireland.

2020 | Presenting my research topic in **IVADO - Digital October.**

2013 | Teaching assistant for **Electronic I**.

2013 | Teaching assistant for **Electronic circuit I**.

2012 | Teaching assistant for **Digital Signal Processing**.

Certificates

2022 | Introduction to Unix Shell, **Calcul Quebec**.

2022 | Massive data analysis with Spark, **Calcul Quebec**.

2022 | Introduction to GPU Programming with Python, **Calcul Quebec**.

2019 | 4th **IVADO/MILA** SCHOOL IN DEEP LEARNING.

2018 | DEEP LEARNING WITH PYTORCH, **Udemy**.

2018 | LTE OPTIMIZATION OF RADIO NETWROK (3GPP), **TechCom**.

2017 | LTE COUNTER AND KPI (ERICSSON), **TechCom**.

2017 | LTE RADIO NETWORK PLANNING, **TechCom**.

Strengths

Hard-working, Decision Making, Ability to Work Under Pressure, Motivator & Leader Conflict Resolution, Time Management, Target oriented, Flexibility, and Adaptability